Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/24/2014
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	RNA Deep Sequencing and Metabolomic Profiling of Microgravity-Induced Regulation of the Host-Pathogen Interaction: An Integrated Systems Approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	Cheryl.Nickerson@asu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	480-727-7520
Organization Name:	Arizona State University		
PI Address 1:	Center for Infectious Diseases and Vaccinology/The Biodesign Institute		
PI Address 2:	1001 S McAllister Avenue		
PI Web Page:	https://		
City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/01/2013	**End Date:**	06/30/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aronow, Bruce (Cincinnati Children's Hospital) Khalpey, Zain (University of Arizona) Ott, Mark (NASA Johnson Space Center) Sams, Clarence (NASA Johnson Space Center) Barrila, Jennifer (CoPI/ Arizona State University)		
Grant/Contract No.:	NNX13AM01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A major challenge in mitigating infectious disease risks during spaceflight is to understand, model, and integrate the combined action of cellular, molecular, and biochemical networks in the host that potentiate transition to disease in response to infection by pathogens cultured in microgravity. This is a critical issue to address, as spaceflight negatively impacts crew immune function and alters microbial virulence, gene expression, and antimicrobial resistance, strongly suggesting an increased risk for in-flight infectious disease. Herein, we propose to apply our 3-D intestinal co-culture models as human surrogates to characterize their transcriptomic (via RNA-Seq), metabolomic (via mass spectrometry), and inflammatory responses to infection with spaceflight analogue and control-cultured Salmonella typhimurium. To facilitate a practical integration of these systems, the combined analysis proposed in this study will focus on the metabolomic subset associated with host oxidative stress, redox, and inflammatory responses. These factors are of key importance in the infection process and may contribute to reported abnormalities and dysfunction in the crews' immune system during flight. We will use systems modeling approaches based on both quantitative gene transcript and biochemical metabolite correlation and all of their known/inferable relationships between genes, proteins, pathways, and biochemistry that will allow us to form new bridges and a general framework for making these linkages between gene expression (RNA-Seq) and biochemical (metabolomic) data. This approach will allow for new insights into relationships between redox functions, oxidative stress, inflammation, and infection that either approach alone would not be able to achieve. By exploring interconnections between these systems over different kinetic timepoints before and after infection, this systematic approach will provide an unparalleled level of sensitivity and resolution of the dynamics of the host response to a microgravity-analogue cultured pathogen, which may lead to identification of novel infection mechanisms and strategies for prevention and control.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform to 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	A change in lab personnel resulted in an approximate three month delay in the start of this project; however, this has been successfully accomplished and we have made significant progress in achieving the goals of this proposal including: • Successfully trained new team members in the development and application of the 3-D co-culture intestinal models for the proposed infection studies. • Completed additional phenotypic validation of the physiological relevance of the 3-D co-culture model. • Optimized conditions for 3-D cell culture infection with LSMMG and control grown Salmonella. • Initial sample sets isolated for downstream –omics analysis.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Crabbé A, Ledesma MA, Nickerson CA. "Mimicking the host and its microenvironment in vitro for studying mucosal infections by Pseudomonas aeruginosa." Pathogens and Disease. 2014 Jun;71(1):1-19. http://dx.doi.org/10.1111/2049-632X.12180 ; PubMed PMID: 24737619 , Jun-2014
Articles in Peer-reviewed Journals	Crabbe A, Nielsen-Preiss S, Woolley CM, Barrila J, Buchanan K, McCracken J, Inglis DO, Searles SC, Nelman-Gonzalez MA, Ott CM, Wilson JW, Pierson DL, Stefanyshyn-Piper HM, Human LE, Nickerson CA. "Spaceflight enhances cell aggregation and random budding in Candida albicans." PLoS One. 2013;8(12): e80677. Published 04 Dec 2013. http://dx.doi.org/10.1371/journal.pone.0080677 ; PMID: 24324620 , Dec-2013